

Staff-Maintainable Local Digital: Handoffs That Outlast a Clinic Semester

Independent research note (portfolio writeup; not peer-reviewed)

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Independent personal writeup; not IU-endorsed or lab output

Abstract

Local organizations often need digital tools their staff can maintain without a developer on call. Clinic sprints that optimize for the demo audience leave partners with undocumented ETL, inaccessible content chunks, and surprise dependencies when the semester ends. This independent writeup frames staff-maintainable local digital work as the operating requirement: accessible by default, honest about scope, and explicit about renewal ownership. Approach: synthesize public-safe project summaries and digital-inclusion principles from the local-digital repository covering ServeIT Clinic builds for Bloomington-area partners (Accessibility Council, Middle Way House, Indiana Phenology, Bloomington Chamber civic reporting). Findings are qualitative process patterns drawn from those notes, not partner performance metrics or invented outcome percentages. Limits: documentation reflects clinic sprint scope as recorded in public project notes; partner-owned data and live URLs stay with handoffs; this PDF is a portfolio-style research note, not peer-reviewed research and not official IU or clinic policy.

1 Introduction

A nonprofit website or civic dashboard that only a student developer can fix has already failed its handoff.

Situation. The concrete setting is clinic work through ServeIT at IU Luddy: community partners need sites, pipelines, and reporting surfaces their own staff can edit, refresh, and escalate after the semester ends and the student team leaves [4].

Gap. Many student builds optimize for a working demo and a gradebook deadline. Accessibility becomes a pre-launch scramble. Data jobs live in a personal account. Documentation describes what the builder did, not what the next maintainer must do. Partners discover those gaps in production. WCAG 2.1 baselines and clinic accessibility guides exist [1, 2]; what this note adds is an operating frame that treats staff maintainability as the acceptance test, not a soft preference after the demo.

Claims. Stated as claims, not empirical results:

1. Local community digital work should be staff-maintainable first: accessible by default, honest about scope, and free of surprise dependencies.
2. That frame is already grounded in four public-safe ServeIT project summaries indexed in local-digital (not new field experiments).
3. Jade’s independent writeup and indexing notes remain separate from partner-owned content and from Serve-AI / ServeIT materials that carry their own

licenses and credit rules.

Section 2 situates related practice; Section 3 describes the frame and projects; Section 4 reports qualitative patterns; Section 5 states limits; Section 6 concludes.

2 Related Work

Clinic accessibility baselines. ServeIT public extension guides target WCAG 2.1 Level AA for clinic builds, with Cascade WCMS and IU Rivet composition patterns [1, 2]. Those guides are the checklist home; local-digital indexes community projects and process notes that do not belong only in accessibility runbooks.

Civic tech and nonprofit handoffs. The project docs emphasize maintainability for communications and admin staff: documented ETL schedules, clear chart semantics, crisis paths above the fold, and chunk-based editing without developer intervention. That is adjacent to broader civic-tech conversations about sustainable public-interest software. This note stays inside documented clinic sprint scope rather than surveying the whole field.

Public-interest AI constraints. When clinic work touches generative AI (Serve-AI / PIT-UN track), the operating frame requires human review before publish, disclosure where appropriate, and no feeding partner PII into external models without consent and clinic policy sign-off [3]. That constraint is part of digital inclusion here, not a separate ethics appendix.

3 Approach

What was analyzed: public project summaries and the digital-inclusion operating frame in the local-digital repository. What was not analyzed: private partner data, credentials, raw dumps, or participant records (explicitly kept out of the public repo).

Four documented clinic projects supply concrete context:

- City of Bloomington Accessibility Council (2024). First clinic client. Accessible page structure aligned with WCAG baselines; Cascade / Rivet composition; documentation for city staff maintainers. City-owned content and deployment URLs stay with the City handoff.
- Middle Way House (2024–2025). Survivor-serving organization. Accessible, responsive web surfaces; staff-maintainable content blocks; critical help paths kept obvious above the fold; security-conscious forms and external links. Live site content belongs to the partner.
- Indiana Phenology (Jan–May 2025). Citizen-science observation ingestion. Python ETL from CSV/XML into PostgreSQL; WCAG-aligned staff workflows. Handoff bar: partner staff can run or schedule ingestion, spot-check failed rows, and pull standard reports without SSH access; failures must log clearly enough for a non-developer to escalate. Raw exports and production credentials stay with the partner.
- Bloomington Chamber civic reporting (Aug–Dec 2025). React / D3 dashboards with a Node.js back end for municipal service-request patterns. Handoff bar: Chamber staff or designated partners can interpret dashboards and refresh data on a documented schedule; chart semantics and filters should read clearly without a developer in the room. Partner-owned data and live deployment URLs stay with the handoff.

3.1 Design Principles

Drawn from the repository’s digital-inclusion frame:

- Staff-maintainable first. If only a student can publish a fix, the handoff failed. Prefer platforms and patterns the partner’s communications or admin staff can edit (Cascade WCMS chunks, documented ETL scripts, clear README handoffs).
- Accessible by default. Keyboard paths, screen reader checks, mixed-language markup, and contrast are part of the build, not a separate audit phase.
- Honest about scope. Document what shipped, what is stubbed, and what needs a developer next semester. Partners and future clinic teams should not discover gaps in production.

- No surprise dependencies. Name APIs, accounts, and renewal owners in the handoff. Avoid hidden API keys, personal accounts, or one-off hosting the partner cannot renew.
- Community data stays community-scoped. Organizational data only in public docs; credentials and raw participant records stay out.

4 Findings

Observed patterns from the public project notes (qualitative; no invented metrics):

1. Handoff bar is the real acceptance test. Across projects, the stated success condition is staff operation (edit chunks, run or schedule ingestion, interpret dashboards), not merely a passing demo. The Accessibility Council sprint was framed as first proof that clinic builds could ship outside the gradebook: real residents, real maintainers, real handoff.
2. Accessibility and maintainability travel together. Cascade / Rivet chunk composition and WCAG baselines appear as first-class build requirements, with detailed runbooks pointed to serveit-accessibility rather than duplicated in local-digital.
3. Data pipelines need human-readable failure modes. Phenology notes require clear logging so non-developers can escalate; silent ETL failure fails the inclusion frame. Staff should spot-check failed rows without SSH.
4. Crisis and civic surfaces need readable semantics under stress. Middle Way House notes keep critical help paths above the fold; Chamber notes require chart semantics and filters that read without a developer in the room.
5. Public docs describe scope, not partner product launches. Each project note states that live URLs, brands, and partner-owned data remain with the handoff. This writeup inherits that boundary.

Evaluation here is documentary: consistency of the operating frame against the four summaries. No A/B tests, partner satisfaction percentages, or traffic claims appear in the source docs, so none are claimed here.

5 Discussion and Limitations

Readers should not conclude that these clinic builds are comprehensively evaluated outcomes research, that Indiana University or ServeIT Clinic endorse this PDF as official research, or that partner organizations have approved this writeup as a product announcement. Limitations: single-author synthesis of already-public project notes; partner deployment details intentionally omitted; clinic sprint periods bound what “shipped” means;

ServeIT / Serve-AI institutional materials remain under their own credit and license rules (CC BY-SA for Serve-AI-related guides). Visual consistency with IU Rivet on the personal site does not transfer ownership of this content to IU. ProHealth / lab clinical work is out of scope and not discussed.

6 Conclusion

Local digital work that outlasts a semester is a maintenance problem before it is a features problem. This independent writeup restates a staff-maintainable operating frame, anchors it in four public ServeIT project summaries, and keeps partner data and institutional licensing boundaries intact. The contribution is process clarity for builders and future clinic teams, not peer-reviewed findings and not IU research output.

Acknowledgments

Independent personal writeup synthesizing public notes in local-digital. Drafting and structure were AI-assisted; accountability remains with Jade Zhao. Clinic context credits ServeIT Clinic (IU Luddy). Where accessibility guides derive from Serve-AI / ServeIT practice, those materials are CC BY-SA 4.0 (credit Serve-AI / ServeIT Clinic). This PDF does not claim sole ownership of Serve-AI upstream materials and is not official IU research.

References

- [1] W3C Web Accessibility Initiative, Web Content Accessibility Guidelines (WCAG) 2.1, W3C Recommendation, 5 June 2018. <https://www.w3.org/TR/WCAG21/>.
- [2] J. Zhao, ServeIT Accessibility (public clinic guides), GitHub repository zhao-langxi/serveit-accessibility. <https://github.com/zhao-langxi/serveit-accessibility>.
- [3] Serve-AI project page (ServeIT Clinic / PIT-UN track). <https://serveit.luddy.indiana.edu/serve-ai/index.html>.
- [4] ServeIT Clinic, Luddy School of Informatics, Computing, and Engineering, Indiana University. <https://serveit.luddy.indiana.edu/>.